

Multiple Choice (10 marks)

1. The model obtained by applying linear regression on the identified subset of features may differ from the model obtained at the end of the process of identifying the subset during
 - (a) Best-subset selection
 - (b) Forward stepwise selection
 - (c) Forward stage wise selection
 - (d) All of the above
2. Statement 1| Industrial-scale neural networks are normally trained on CPUs, not GPUs. Statement 2| The ResNet-50 model has over 1 billion parameters.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
3. Averaging the output of multiple decision trees helps __.
 - (a) Increase bias
 - (b) Decrease bias
 - (c) Increase variance
 - (d) Decrease variance
4. High entropy means that the partitions in classification are
 - (a) pure
 - (b) not pure
 - (c) useful
 - (d) useless
5. Statement 1| Overfitting is more likely when the set of training data is small. Statement 2| Overfitting is more likely when the hypothesis space is small.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: The Penn Treebank is primarily used for sentiment analysis rather than for language modeling.
7. True or False: Increasing the number of training examples indefinitely will result in higher variance for the model's predictions.
8. True or False: K-means clustering, partitioning based clustering, and grid based clustering are all examples of spatial clustering algorithms.
9. True or False: The cost of one gradient descent update given the gradient is $O(ND)$, where N is the number of data points.
10. True or False: An increase in training loss with more epochs may suggest that regularization is too low, leading to overfitting.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to find the minimum difference between any two elements in a given array.
12. Write a python function to check whether every odd index contains odd numbers of a given list.

End of Paper